R22

20/11/2025

Code No.: 22CSC21

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.E. (CSE, IT & AI&DS) V Sem (Main/Backlog) Examination November 2025

Software Engineering

Time: 3 Hours **Max Marks: 60**

Note: "Answer **ALL** questions from **Part-A** and **Part-B** (including **internal choices**) at one place and in the same order as in the question paper."

Part - A
(5Q X 2M = 10 Marks)

			M	CO	BT
1		Suggest two advantages and two disadvantages of the approach to process maturity that is embodied in the Software Engineering Institute (SEI's) Capability Maturity Framework.	(2)	1	1
2		What is the significance of Scrum?	(2)	2	1
3		What is Problem-Based Estimation?	(2)	3	1
4		Differentiate between Black-Box Testing and White-Box Testing strategies.	(2)	4	1
5		Who should perform the validation test—the software developer or the software user? Justify your answer.	(2)	5	1

Part - B
(5Q X 10M = 50 Marks)

			M	CO	BT
6	(a)	Umbrella activities occur throughout the software process. Do you think they are applied evenly across the process or are concentrated in one or more framework activities?	(5)	1	2
	(b)	Explain the Management myths. State the reality that accompanies the Practitioner's myth.	(5)	1	2
		(OR)			
7	(a)	As you move outward along the spiral process flow, what can you say about the software that is being developed or maintained? Explain your answer.	(5)	1	2
	(b)	Compare the RAD Model with Prototyping Model.	(5)	1	2
8	(a)	Explain the features, and applications of Feature Driven Development (FDD) agile model.	(5)	2	2
	(b)	Describe the XP concepts of refactoring and pair programming in your own words.	(5)	2	3
		(OR)			
9	(a)	Suggest why it is important to make a distinction between developing the user requirements and developing system requirements in the requirements engineering process.	(5)	2	2
	(b)	Develop a requirements gathering "kit." The kit should include a set of guidelines for conducting a requirements gathering meeting and materials that can be used to facilitate the creation of lists and any other items that might help in defining requirements.	(5)	2	3
10	(a)	How can you say that performance is important consideration during Planning? Justify your answer.	(5)	3	2
	(b)	Describe the difference between risk components and risk drivers.	(5)	3	2

(OR)

11	(a)	Describe the various Software Sizing methods.	(5)	3	2
	(b)	Assume that you are the project manager for a company that builds software for household robots. You have been contracted to build the software for a robot that mows the lawn for a homeowner. Do a functional decomposition of the robot software. Estimate the size of each function in LOC. Assuming that your organization produces 450 LOC/pm with a burdened labor rate of $7000 per person-month, estimate the effort and cost required to build the software using the LOC-based estimation technique.	(5)	3	3
12	(a)	Explain the design Guidelines.	(5)	4	2
	(b)	What is Modular Design? Why it is needed? Explain with one example.	(5)	4	2

(OR)

13	(a)	Describe the User Interface Design Principles and Guidelines. Illustrate the evaluation cycle of user interfaces for a system of your choice.	(5)	4	2
	(b)	Is it reasonable to say that problem domain components should never exhibit external coupling? If you agree, what types of components would exhibit external coupling?	(5)	4	2
14	(a)	Explain how the various types of loops are tested.	(5)	5	2
	(b)	What is cyclomatic complexity? What are the ways to compute it? Give the steps to select the path in data flow testing.	(5)	5	2

(OR)

15	(a)	Describe the McCall's quality factors.	(5)	5	2
	(b)	A formal technical review is effective only if everyone has prepared in advance. How do you recognize a review participant who has not prepared? What do you do if you're the review leader?	(5)	5	2

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